SAMPLE PAPER 1
Level 2 Functional Skills Mathematics

City & Guilds
A City & Guilds Group Business

Duration: 25 minutes
Total marks: 15

SECTION 1 – CALCULATOR NOT PERMITTED
VERSION 1.0

MODEL SOLUTION SCORE
15 / 15 marks (100% Pass)
25 minutes · Verified by MFS

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

You should have the following for this assessment:

- a pen with black or blue ink
- a pencil (for diagrams, graphs and charts only)
- an eraser
- a 30cm ruler.



You must NOT use a protractor.
You must NOT use a calculator for Section 1.

General instructions

- Read through each question carefully.
- Write all your answers in this booklet.
- Check your calculations and check that your answers make sense.

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SECTION 1 – CALCULATOR NOT PERMITTED

There are **15** marks available in this section.

You should check all your work as you go along.

You must **not** use a calculator in this section.



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Q1

What is $\frac{2}{3}$ as a percentage? Give your answer rounded to two decimal places.

[BI]

66.67 %

(1 mark)

Method: 2 ÷ 3 = 0.6666... × 100 = 66.666...%

Round to 2 d.p.: 3rd decimal digit is 6 (≥ 5), round up --> 66.67%

Q2

What is 14% of 200?

[BI]

28

(1 mark)

10% of 200 = 20 · 1% of 200 = 2 --> 4% = 8

14% = 20 + 8 = 28 (Check: 0.14 × 200 = 28)

Q3

$2\frac{1}{2} + 3\frac{3}{4} =$

(tick one box)

A $5\frac{1}{4}$ ☐

B $5\frac{4}{6}$ ☐

C $6\frac{1}{4}$ ☑

D $6\frac{4}{6}$ ☐

Whole numbers: 2 + 3 = 5

Fractions: 1/2 + 3/4 = 2/4 + 3/4 = 5/4 = 1 1/4

Total: 5 + 1 1/4 = 6 1/4 --> Option C

[BI]

(1 mark)

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Q4

What is 75 as a fraction of 125? Give your answer in its simplest form.

[B1]

Simplify by Highest Common Factor (HCF = 25):

75 / 125 = (75 ÷ 25) / (125 ÷ 25) = 3 / 5

$\frac{3}{5}$

(1 mark)

Q5

$\frac{1}{2} - \frac{2}{7} =$

Give your answer in its simplest form.

[B1]

Common Denominator (LCM of 2 and 7 = 14):

1/2 = 7/14 2/7 = 4/14

7/14 - 4/14 = 3 / 14

$\frac{3}{14}$

(1 mark)

Q6

$(8 + 2 \times 6)^2 =$

BIDMAS: Brackets first, multiplication inside:

$(8 + 12)^2 = 20^2 = 20 \times 20 = 400$

[B1]

400

(1 mark)

Q7

What is the value of $3ab$ when $a = 5$ and $b = 6$?

[B1]

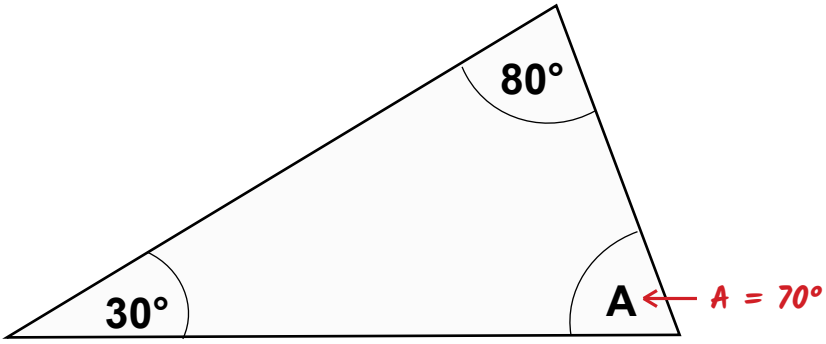
Substitute values: 3 x a x b = 3 x 5 x 6 = 15 x 6 = 90

90

(1 mark)

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Q8



Calculate the size of angle A.

[B1]

Angle Rule: Angles in a triangle sum to 180°:

180° − (80° + 30°) = 180° − 110° = 70°

70 °

(1 mark)

Q9

900 + 1500 ÷ 300 =

[B1]

BIDMAS: Division before Addition (1500 ÷ 300 = 5):

900 + 5 = 905

905

(1 mark)

Q10

147.206 – 95.438 =

[B1]

Column Subtraction: 147.206 − 95.438 = 51.768

51.768

(1 mark)

Q11 A car can travel 480 miles on a full tank of petrol. The tank holds 60 litres.
The fuel gauge shows there are 15 litres left in the tank.

How many more miles can the car travel before it runs out of petrol?

[B1]

Fuel Rate: 480 miles ÷ 60 litres = 8 miles per litre

Remaining: 15 litres × 8 miles/litre = 120 miles

Check: 15/60 = 1/4 tank --> 480 ÷ 4 = 120 miles

120 miles

(1 mark)



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Q12 The probability that a salesperson will get an order from a visit to a customer is $\frac{1}{4}$

She has 2 visits tomorrow.

What is the probability that she will get orders from **both** visits tomorrow?

Give your answer as a fraction in its simplest form.

Independent Events Rule: P(A and B) = P(A) × P(B)

P(Visit 1 and Visit 2) = 1/4 × 1/4 = 1 / 16

[B1]

$\frac{1}{16}$

(1 mark)

Q13 This table shows the change in the number of employees in different departments of a company compared to last year.

Department	Change compared to last year
Admin	-1
Design	0
Production	+4
Packing	+2
Warehouse	-1
Marketing	-3

What is the **total** change in the number of employees compared to last year?

(tick one box)

A 1 fewer ☐

B 1 more ☑

C 11 fewer ☐

D 11 more ☐

[B1]

(1 mark)

Sum Changes:

Positives: +4 + 2 = +6

Negatives: -1 - 1 - 3 = -5

Total: +6 - 5 = +1 (1 more) --> Option B

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Q14 A manager wants to give a pay rise to everyone who is paid less than the average salary.
This table shows the annual salaries of the employees in the company.

Ordered (£k):
15.5 (MT)
16.0 (AJ)
18.5 (RD)
20.0 (JR)
[22.0] SW
23, 23, 23, 36

Median = £22k
4 staff < £22k

Employee	Salary (in £ thousands)	
AJ	16	☑
TM	23	☐
WF	23	☐
SW	22	☐
MT	15.5	☑
RD	18.5	☑
JR	20	☑
LS	23	☐
PB	36	☐

Tick all of the employees who are paid less than the median salary.

[B1]

(1 mark)

Q15
The distance between two villages on a map measures 6 centimetres.

The map has a scale 1:25 000

What is the actual distance between the two villages in **kilometres**?

[B1]

1.5 km

(1 mark)

Map Scale: 1 cm on map = 25,000 cm in reality
6 cm on map = 6 × 25,000 = 150,000 cm
Convert to metres: 150,000 ÷ 100 = 1,500 m
Convert to kilometres: 1,500 ÷ 1,000 = 1.5 km

End of Section 1.
When you have finished you MUST hand this booklet in
to the invigilator before you pick up your calculator to start Section 2.

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SAMPLE PAPER 1
Level 2 Functional Skills Mathematics

City & Guilds
A City & Guilds Group Business

Duration: 1 hour 20 minutes
Total marks: 45

SECTION 2 - CALCULATOR PERMITTED
VERSION 1.0

MODEL SOLUTION SCORE
45 / 45 marks (100% Pass)
1 hr 20 mins · Verified by MFS

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

• If you have used any additional answer sheets write the number of additional sheets in this box. ☐
• Please ensure that you **staple** additional answer sheets to the **back** of this booklet, clearly labelling them with your full name, enrolment number, centre number and date in BLOCK CAPITALS.
• You must use a black or blue pen. You may use a pencil for charts and diagrams.
***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

You should have the following for this assessment

- a pen with black or blue ink.
- a pencil (for diagrams, graphs and charts only)
- an eraser
- a 30cm ruler.



You may use a calculator for Section 2.

You must NOT use a protractor.

General instructions

- Read through each question carefully.
- Show your working out (where required).
- Write all your working out and answers in this booklet.
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SECTION 2 - CALCULATOR PERMITTED

There are **45** marks in this section.

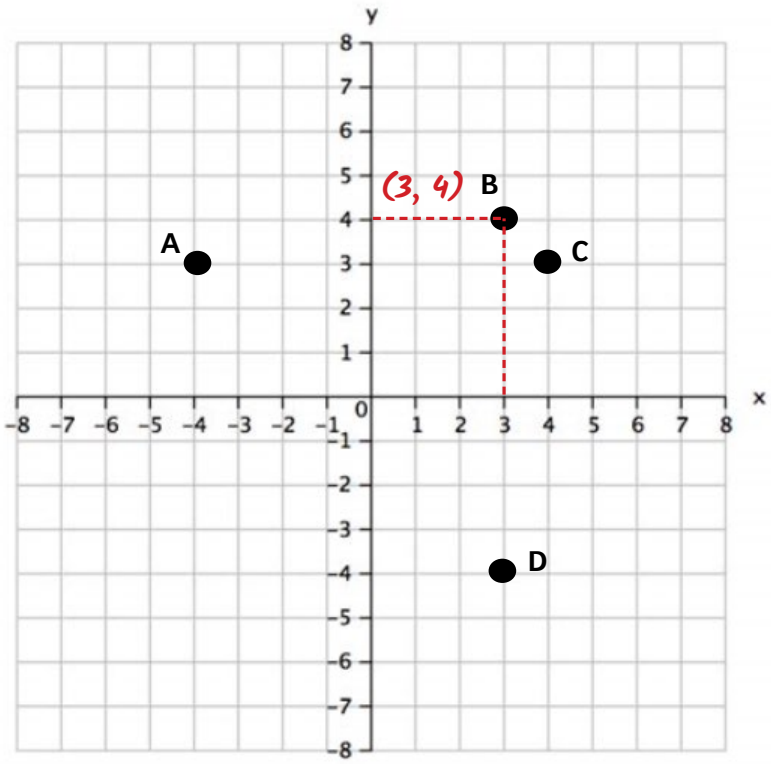
You should check all your work as you go along.

You may use a calculator.



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Q1



Which point is at (3,4)?

(tick one box)

A. Point A ☐

B. Point B ☑

C. Point C ☐

D. Point D ☐

Coordinate Rule: (x, y) = (3, 4)

- From (0,0), move 3 units right on x-axis
- Move 4 units up on y-axis

--> Identifies Point B

[B1]

(1 mark)

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Q2

1 gallon = 4.546 litres

10 litres in gallons is approximately

(tick one box)

A. 0.45 gallons ☐

B. 2.2 gallons ☑

C. 45.5 gallons ☐

D. 22 gallons ☐

Calculation: 10 L ÷ 4.546 L/gal = 2.1997... gal
Round to 1 d.p. = 2.2 gallons --> Option B

[B1]

(1 mark)

Q3

155	125	145	90	125	150	155
90	100	125	178	95	125	180

What is the mode of these numbers?

[B1]

Mode = Most frequent value: 125 appears 4 times (highest frequency)

125

(1 mark)

Q4

Which one of the following lists is in increasing order?

(tick one box)

A. 0.1013 0.0827 0.0095 ☐

B. 0.1013 0.0095 0.0827 ☐

C. 0.0095 0.1013 0.0827 ☐

D. 0.0095 0.0827 0.1013 ☑

Compare thousandths place values:
0.0095 < 0.0827 < 0.1013
--> Option D is in ascending order

[B1]

(1 mark)

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Q5

The surface area of a sphere is $4\pi r^2$

A sphere has a radius (***r***) that measures 3cm

Use $\pi = 3.142$
or $\pi = \frac{22}{7}$

Work out the surface area of the sphere to the nearest cm^2.

A. $15cm^2$ ☐

B. $22cm^2$ ☐

C. $113cm^2$ ☑

D. $1421cm^2$ ☐

Formula: Surface Area = 4 × pi × r^2

Substitute r = 3 cm, pi = 3.142:

Area = 4 × 3.142 × 3^2 = 113.112 cm^2

To nearest cm^2 = 113 cm^2 --> Option C

[B1]

(1 mark)

Q6 A man is going to New York for work. He wants to book a hotel online.

A friend says
'Remember the booking website will show the price in dollars. It will actually cost **more pounds** than the price shown, because of the exchange rate.'

The man checks the exchange rate because he thinks his friend is wrong. He thinks that the number of pounds will be **less** than the number of dollars shown.

Exchange rate **£1 = $1.24**

Who is right, the man or his friend?
Explain your answer.

[B1]

Explanation

Decision: The man is RIGHT (the friend is wrong).

Explanation: The British Pound is worth more than the US Dollar (£1 = $1.24).

Converting dollars to pounds divides by 1.24, so the price in £ is less.

(1 mark)

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Q7 A newspaper report says that a company made £700,000 profit last year.
It says this was 12% more than the year before.

Work out how much profit the company made the year before.

Show all your working

Step 1: Reverse Percentage Multiplier:

£700,000 represents 100% + 12% = 112% of previous year:

Multiplier = 1.12 [M1]

Step 2: Divide by multiplier to find original profit:

Original Profit = £700,000 ÷ 1.12 [M1]

Original Profit = £625,000 [A1]

Check: 12% of £625,000 = £75,000 · £625,000 + £75,000 = £700,000 [OK]

Profit £ 625,000

(3 marks)

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Q8

Income tax

Everyone can earn a certain amount of money without paying tax. This is called a Personal Allowance. They must pay tax on any earnings over this allowance.

Income tax Personal Allowance, 2018/2019	£11 850

This formula gives the amount of Income tax a person pays in a year

$$T = 0.2\ (y - p)$$

where T = income tax **for the year**
y = money earned per **year**
p = Personal Allowance

A caterer earns £1 375 per **month**.

How much income tax will she pay for the **year**?

Show all your working.

Step 1: Calculate annual gross income (y):

y = £1,375 per month × 12 months = £16,500 per year [M1]

Step 2: Calculate taxable earnings (over Personal Allowance p = £11,850):

Taxable amount = y − p = £16,500 − £11,850 = £4,650 [M1]

Step 3: Apply income tax formula: T = 0.2 × (y − p):

T = 0.2 × £4,650 = £930.00 [M1]

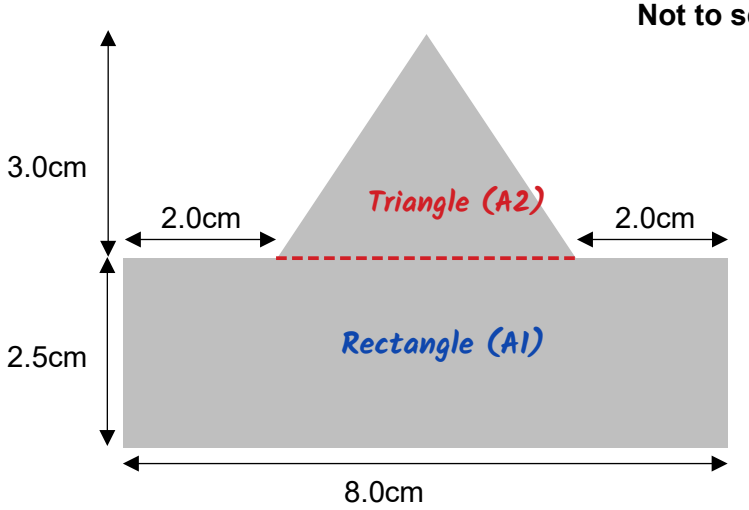
£ **930.00** [A1]

(4 marks)

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Q9 A worker has to set a machine to cut this shape from a piece of metal.

Not to scale



What is the area of the shape?

Show all your working.

Step 1: Area of Rectangle (A1) = base × height:

Area (A1) = 8.0 cm × 2.5 cm = 20.0 cm² [M1]

Step 2: Base = 8.0 - 2.0 - 2.0 = 4.0 cm; Height = 3.0 cm:

Area (A2) = 1/2 × 4.0 cm × 3.0 cm = 6.0 cm² [M1]

Step 3: Total Area = 20.0 cm² + 6.0 cm² = 26.0 cm² [A1]

26 cm²

(4 marks)

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Q10 A photographer increases the price he charges to print photographs.
He wants to know if this affects his sales.

Last week, before the price increase, the average number of photos ordered was 12.

This week customers ordered:

Photos ordered	Number of customers	Midpoint (x)	f x x
1 - 10	26	5.5	143.0
11 - 20	14	15.5	217.0
21 - 30	6	25.5	153.0
31 - 40	4	35.5	142.0
41 - 50	0	45.5	0.0
51 - 60	0	55.5	0.0

Does the price increase seem to have had an effect on the number of prints ordered per customer? Explain your answer. Include calculations to support your decision.

Decision (yes/no) Yes

Explanation and supporting calculations

Step 1: Calculate total frequency and total photos (Total fx):

Total f = 26 + 14 + 6 + 4 = 50 customers

Total fx = 143 + 217 + 153 + 142 = 655 photos [M2]

Step 2: Estimated Mean = Total fx ÷ Total f:

Estimated Mean = 655 ÷ 50 = 13.1 photos per order [A1]

Step 3: Comparison & Decision:

Before price rise, mean was 12 photos. New mean is 13.1 photos.

Average increased by 1.1 photos, showing that sales did not drop. [A1]

(4 marks)

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Q11 This table shows how much a garage pays its staff.

Pay rates	
Working day	**Rate**
Monday to Friday	Normal rate
Saturday or Sunday	1¼ x normal rate

Last week, a mechanic worked 7½ hours each day from Monday to Saturday. She did not work on Sunday.

Her normal rate of pay is £10.80 an hour.

Work out her **total** pay for last week.

Show all your working

Step 1: Monday to Friday Pay (5 days @ 7.5 hrs):

5 days × 7.5 hrs/day = 37.5 hours • 37.5 hrs × £10.80 = £405.00 [M1]

Step 2: Saturday Pay (1.25 × normal rate):

Saturday rate = 1.25 × £10.80 = £13.50/hr • 7.5 hrs × £13.50 = £101.25 [M1]

Step 3: Total Pay = £405.00 + £101.25 = £506.25

£ 506.25 [A1]

Use approximation to check your answer.

Do your check here

Approximation Check: Mon-Fri: 5 × 7 hrs × £10 = £350

Saturday: 1 × 8 hrs × £13 = £104 --> £350 + £104 = £454 [OK] [B1]

(4 marks)

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Q12 A woman applies for a new job that pays £8.50 a week more (after tax).

She will work 5 days a week and drive to work, as she does in her job now.
The new job is 6 miles further from her house.

Her car travels 8.5 miles per litre of petrol
Petrol costs £1.26 per litre

Will the woman be better off with the new job after she takes the petrol into consideration?

Explain your answer. Include calculations to support your decision.

Decision (yes/no) No

Explanation and supporting calculations

Step 1: Calculate extra mileage driven per week:

Daily extra = 6 miles each way × 2 = 12 miles round trip

Weekly extra (5 days) = 5 × 12 miles = 60 miles per week [M1]

Step 2: Petrol required and extra fuel cost:

Petrol used = 60 miles ÷ 8.5 miles/litre = 7.0588... litres

Extra fuel = 7.0588 L × £1.26 = £8.894... ≈ £8.89/wk [M1]

Step 3: Comparison with £8.50 pay rise:

Extra pay = £8.50/week vs Extra fuel = £8.89/week

Net Loss = £8.50 - £8.89 = -£0.39 (loses 39p/week). [A2]

(4 marks)

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Q13 Your boss needs you to make some travel arrangements for him.

He will travel to Hull **4 days** every week for the next 6 months (26 weeks).

He needs to arrive at Hull at 8:30am and catch the train home at 5pm each day.

TRAIN TICKET PRICE INFORMATION

TRAINS TO HULL

TICKET TYPE:

DAY RETURN	£8.00
OFF-PEAK* DAY RETURN	£6.20

SEASON TICKETS VALID FOR:

ONE WEEK	£29.60
ONE MONTH	£113.70
ONE YEAR	£1184.00

(Price for season ticket covers all travel while the ticket is valid)

* **OFF-PEAK** tickets are **not** valid for travel between 0700 and 0900 or between 1500 and 1900

Which ticket type do you recommend?

Recommendation

Monthly Season Tickets

Explain your reasons. Include figures or calculations to support your decision.

Explanation and supporting calculations

Constraints: 4 days/wk × 26 wks = 104 travel days. 8:30am arrival requires PEAK tickets.

1. Day Return (Peak): 104 days × £8.00 = £832.00 [M1]

2. Weekly Season: 26 weeks × £29.60 = £769.60 [M1]

3. Monthly Season: 6 months × £113.70 = £682.20 (CHEAPEST) [M1]

Comparison: Monthly (£682.20) is £149.80 cheaper than Day Returns, and £87.40 cheaper than Weekly Season tickets.

Recommendation: Monthly Season Tickets (£682.20 total cost). [A2]

(5 marks)

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Q14 A company has made some changes to the way its employees work.

The manager wants to know if these changes have made any difference to the number of days employees take off work because of illness.

She can't just compare the total days as there are fewer people working in each department after the changes.

She gives you this information about the employees in one department.

Number of days each employee took off sick in the year BEFORE the changes			
14	12	11	8
12	0	15	6
11	3	10	7
0	5	8	10
15	16	14	3

Number of days each employee took off sick in the year AFTER the changes		
12	0	2
11	3	7
14	10	10
3	8	9
8	4	4

Did the changes make any difference to the average number of days that employees took off sick?

Explain your findings to the manager. Show calculations to support your explanation.

Decision (yes/no) Yes

Explanation and supporting calculations

BEFORE changes (20 staff): Total = 180 sick days

Mean Before = 180 ÷ 20 = 9.0 days per employee [M1]

AFTER changes (15 staff): Total = 105 sick days

Mean After = 105 ÷ 15 = 7.0 days per employee [M1]

Comparison: Average sick days dropped from 9.0 to 7.0 days (a 2-day reduction).

Conclusion: Yes, the changes reduced the average number of sick days t [A3]

(5 marks)

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Q15 A café owner wants to know how many cold drinks she is likely to sell next week.

She makes a record of sales of drinks over the last two weeks:

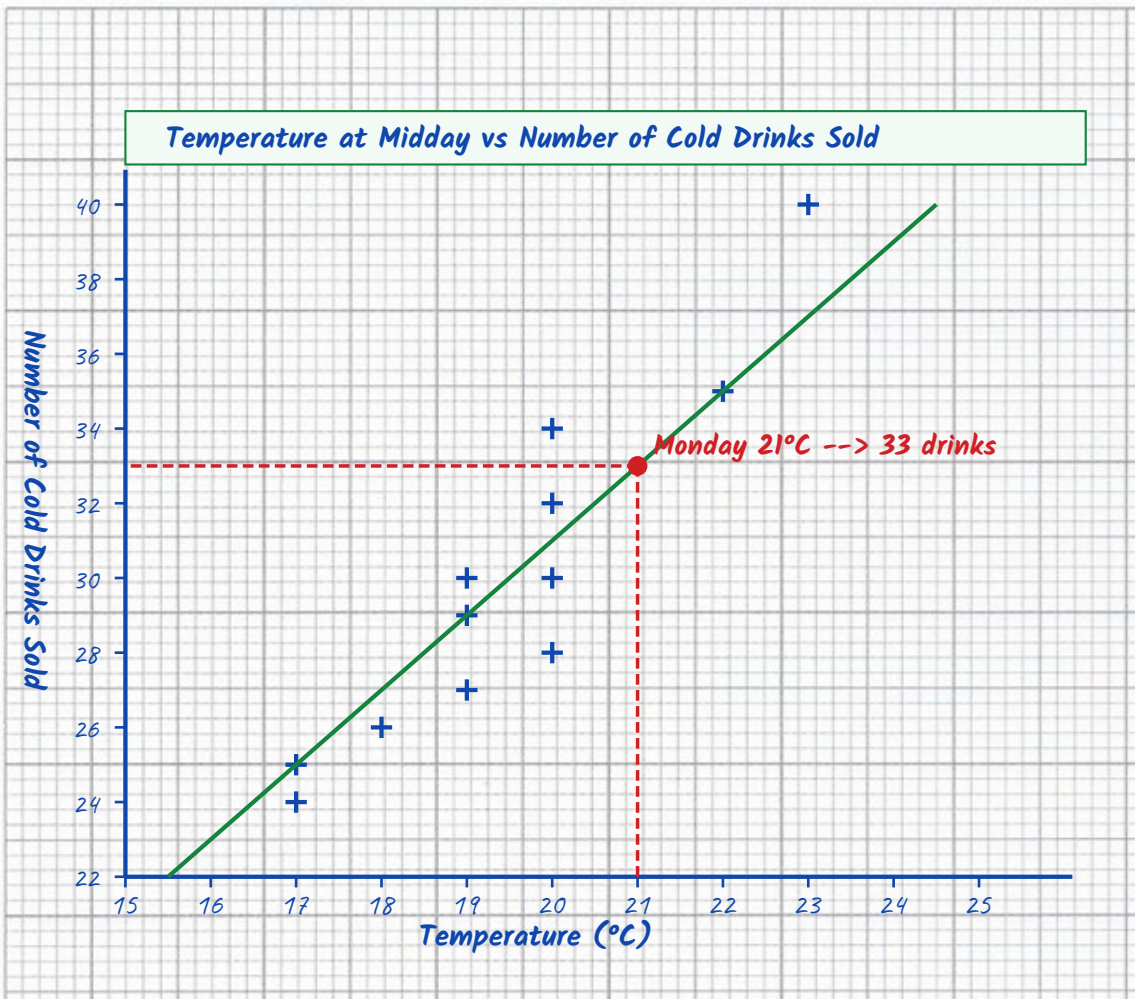
Day	M	T	W	Th	F	S	M	T	W	Th	F	S
Temperature (°C) at midday	**17**	**18**	**17**	**19**	**20**	**20**	**19**	**19**	**22**	**23**	**20**	**20**
Number of cold drinks sold	**24**	**26**	**25**	**30**	**32**	**28**	**27**	**29**	**35**	**40**	**30**	**34**
Number of hot drinks sold	**34**	**36**	**32**	**34**	**27**	**29**	**37**	**39**	**25**	**25**	**28**	**28**

She wants to use this information to see if she can predict the number of **cold** drinks she is likely to sell based on the temperature forecast for a particular day.

Use the graph paper to show clearly the data she has collected in a way that will help her to do this.

Space for working

- Mean Point (Centroid): Mean Temp = 19.5°C, Mean Drinks = 30.0
- Correlation: Strong positive correlation (hotter weather increases cold drinks sold)
- Prediction: Reading 21°C on line of best fit gives 33 cold drinks for Monday



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The weather forecast for next week says it will be 21°C on Monday.

What can you tell the café owner about how many cold drinks the café is likely to sell on Monday?
Show clearly on your graph paper how you found your answer.

Answer 33 cold drinks (accept 32 – 34) [A1]

- Read from 21°C on line of best fit across to drinks axis gives 33 cold drinks.
- Strong positive correlation: higher temperatures correspond to higher cold drink sales.

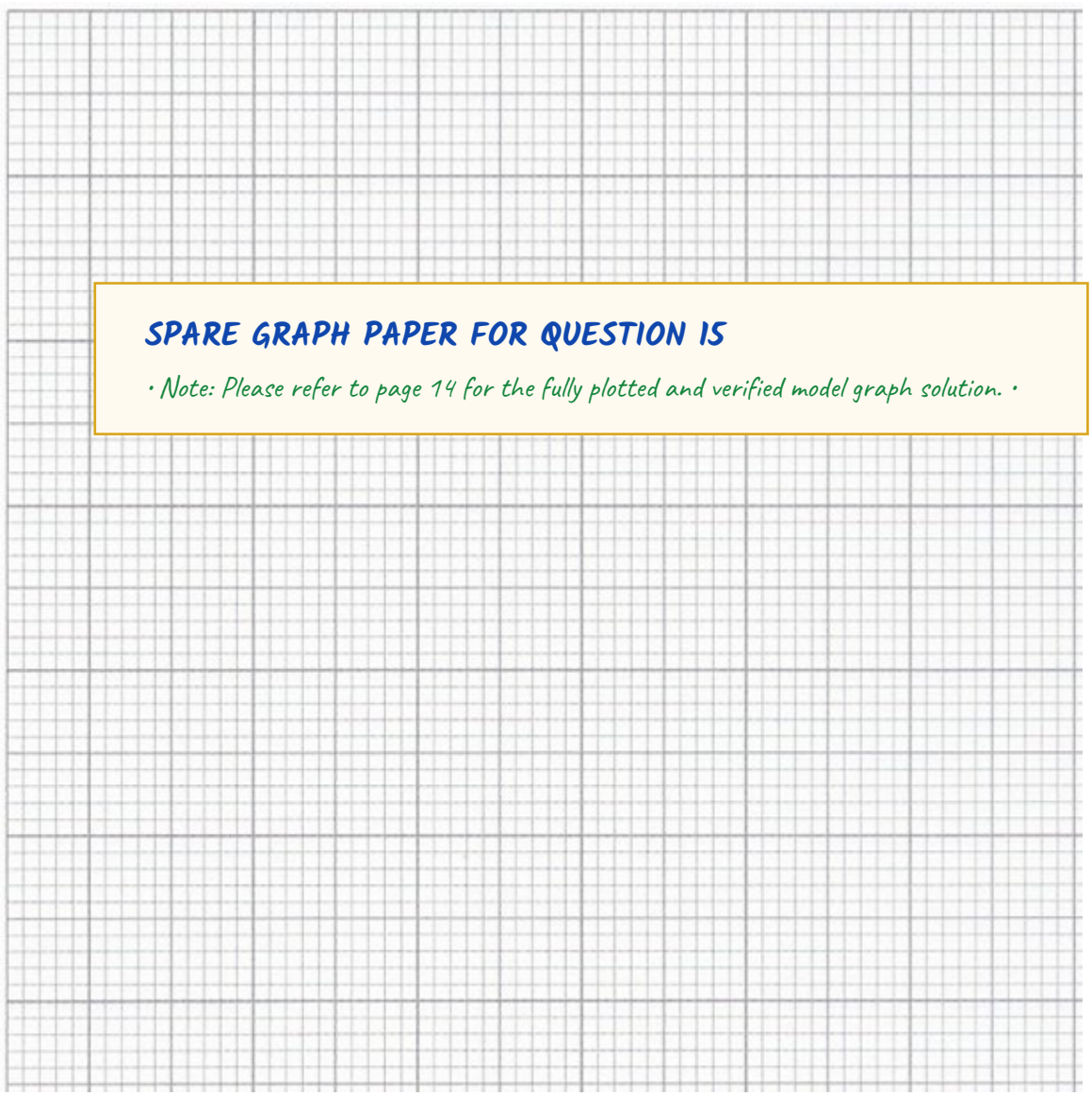
Conclusion: At 21°C on Monday, expected cold drink sales are approximately 33 drinks (interpolated from line of best fit).

(6 marks)

End of Section 2

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Spare graph paper for Question 15



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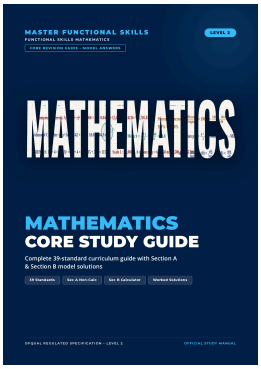
End of assessment

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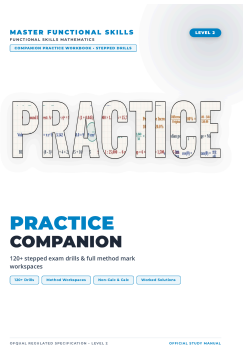


Core Curriculum Manual

Maths Level 2
Core Study Guide

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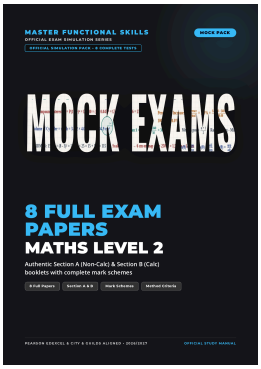


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